ANTIBACTERIAL DRUGS

(Anti-folates)

BIETE LUNDAU LUKE

Mclinpharm, Bpharm, DipPharm

ANTI-FOLATE DRUGS

OVERVIEW

- Folates are very cardinal when it comes to replication or metabolism of bacterial DNA

- Synthesis of folate in bacteria usually begins with the fusion of pteridine and *p-aminobenzoic* acid (PABA) to form **dihydrofolate**

- The enzyme involved in the first process is called **dihydropteroate synthetase**

- Dihydrofolate is then converted to **tetrahydrofolate** by **folate reductase**

- Unlike the bacteria, mammals are unable to synthesize their dihydrofolate and they must obtain it from their diet which once absorbed must be converted to tetrahydrofolate and other active metabolites

- Although folate reductase is found in both microbial and mammalian cells, the affinity for the enzyme in bacteria by drugs like trimethoprim is about 100,000 times greater than its affinity for the mammalian enzymes

Classification of anti-folate drugs

There are two main groups of anti-folate drugs namely;

1. Suphonamides

- Sulfadoxine
- Sulfamethoxazole
- Silver sulfadiazine
- Sulfacetamide

2. Others

- Pyrimethamine
- Trimethopril
- Methotrexate

SULFONAMIDES

Mechanism of action

- Sulfonamides are structural analogues of PABA and therefore, competitively inhibit dihydropteroate synthetase
- This subsequently inhibits the formation of dihydrofolate
- Because of their competitive nature of action of sulfonamides, it is important to appreciate that their effect can be counteracted by the administration of PABA

Indications

- Sulfonamides were the first drugs to have been used in systemic bacterial infections and were once upon a time active against a wide variety of organisms which unfortunately have become resistant over the years

- Primarily used to prevent and treat urinary tract infections

- Sulfadiazine is available in the form of silver sulfadiazine ointment or cream used to prevent or treat burns infections and other superficial or skin infections (Silver ions have antibacterial activity and contribute to its efficacy)

- Sulfacetamide is administered topically to treat blepharitis and conjunctivitis and also active against trachoma, a highly contagious ocular infection caused *Chlamydia trachomatis*

Adverse effects

- Skin rashes which can be mild or progress to a serious life threatening forms such as erythema multiform or Stevens -Johnsons syndrome

- Crystalluria – sulfonamides are converted to inactive compounds by N-acetylation of which the parent drug is more soluble in urine compared to the acetylated metabolite leading to precipitation and finally the formation of urine crystals (The reason why patients taking sulfonamides should consume adequate quantities of water)

- Gastrointestinal reaction, headaches, hepatitis, haematopoetic toxicity

- Haemolytic anaemia especially patients with glucose-6-phosphate dehydrogenase (G6PD) deficiency

TRIMETHOPRIM

Mechanism of action

- Inhibits the folate reductase enzyme which is responsible for conversion of dihydrofolate to tetrahydrofolate

Indications

- Bacteria prostatis and vaginitis as it tends to concentrate in prostate tissue and vagina
- Active against many gram negative bacilli and a few gram positive organisms
- Usually administered together wither sulfamethoxazole i.e sulfamethoxazole + trimethoprim (Cotrimoxzole), common brand being septrin

Adverse effects

- Nausea, vomiting and epigastric pain
- Rashes and skin hypersensitivity
- Hepatitis, thrombocytopaenia, leukopaenia, haematologic disorders

TRIMETHOPRIM-SULFAMETHOXAZOLE (TMP-SMX)

- A fixed dose combination of sulfamethoxazole and trimethoprim (Cotrimoxazole), common brand being septrin

- Have synergistic activity against susceptible organisms

- The combination is in the 5:1 ratio of sulfamethoxazole and trimethoprim respectively which produces 20:1 plasma concentration because trimethoprim has a greater volume of distribution than sulfamethoxazole

Indications of TMP-SMX

- Exhibits bactericidal activity against some organisms that are not susceptible to either drug given alone
- Active against members of the family Enterobactericeae including *E.coli, Klebsiella pneumonie, proteus* species and *Enterobacter species*
- Prevention and treatment of urinary and prostate infections caused by the earlier mentioned organisms though it should not be used as empiric treatment of UTIs in areas where 30% of *E.coli* isolates have proved resistance against it to this combination
- Drug of choice for treatment pneumonia infections caused by *Pneumocystis jiroveci* and Nocardia asteroids which most occur in immune compromised individuals
- Active against some strains of salmonella and Shigella but other strains are resistant

Adverse effects of TMP-SMX

- Adverse effects are similar to those caused by individual drugs
- Can cause megaloblastic anaemia in persons who have low dietary intake of folic acid though this adverse effect is rare

END